

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA5607

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)*

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Dr.

QUESTIONS: 1 (1 X 100 = 100)

Total Marks: 100 Duration: 3hrs

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications. (BL6)
2. Develop a cloud-edge communication model for wireless healthcare monitoring. (BL6)
3. Design a functional architecture for real-time edge data processing in industrial automation. (BL6)
4. Propose a cloud-based wireless framework for smart surveillance systems. (BL6)
5. Design an edge AI workflow for intelligent wireless data filtering. (BL6)
6. Develop a task offloading strategy between edge and cloud in 5G applications. (BL6)
7. Design a low-latency communication architecture for edge-assisted smart city services. (BL6)
8. Propose a scalable edge-cloud framework for wireless sensor networks. (BL6)
9. Design a Cloud Radio Access Network model for dense urban deployment. (BL6)
10. Develop a functional split model between centralized and distributed radio units. (BL6)
11. Design a virtualized wireless access framework for next-generation communication. (BL6)
12. Propose a data flow model for cloud-based radio resource allocation. (BL6)
13. Design a centralized traffic control model using Cloud RAN principles. (BL6)
14. Develop a network management architecture for cloud-assisted wireless systems. (BL6)
15. Design a wireless service framework integrating virtual network functions. (BL6)
- 16. NB-IoT Architecture and Emerging Industrial Applications (BL5)**
17. Design a satellite-assisted communication model using Low Earth Orbit satellite for remote wireless coverage. (BL6)
18. Develop a comparative architecture for LEO, MEO, and GEO backhaul systems. (BL6)

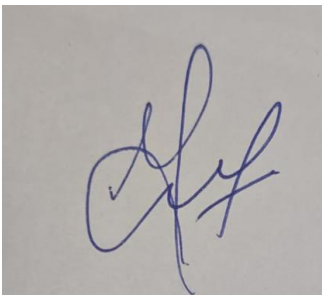
19. Design a spectrum sharing model between LTE and 5G users. (BL6)
20. Propose a frequency allocation strategy for mixed wireless services. (BL6)
21. Design a dynamic spectrum access framework for urban wireless systems. (BL6)
22. Develop a communication architecture combining terrestrial and satellite networks. (BL6)
23. Design a spectrum efficiency improvement model for dense communication zones. (BL6)
24. Propose a channel reuse plan for multi-user wireless communication. (BL6)
25. Design an NB-IoT architecture for smart utility monitoring. (BL6)
26. Develop an LTE-M communication workflow for wearable monitoring systems. (BL6)
27. Design a low-power communication model for battery-operated IoT devices. (BL6)
28. Propose a device communication plan for long-life sensor networks. (BL6)
29. Design a 5G-enabled IoT framework for campus automation. (BL6)
30. Develop an energy-efficient wireless communication strategy for sensor nodes. (BL6)
31. Design an IoT communication flow for periodic environmental sensing. (BL6)
32. Propose a scalable wireless architecture for smart connected devices. (BL6)
33. Design a smart city communication framework for traffic monitoring and surveillance. (BL6)
34. Develop an energy monitoring communication model for public infrastructure. (BL6)
35. Design a smart factory information flow for predictive maintenance. (BL6)
36. Propose a robotic automation communication framework for industrial control. (BL6)
37. Design a remote patient monitoring communication architecture. (BL6)
38. Develop a wireless communication workflow for AI-assisted healthcare diagnosis. (BL6)
39. Design a smart irrigation communication model for crop monitoring. (BL6)
40. Propose a livestock tracking communication architecture using wireless sensors. (BL6)

Code of Conduct:

*I **M.Mukesh Balaji (192412156)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



NB-IoT Architecture and Emerging Industrial Applications

1. Understanding of the Topic

Narrowband Internet of Things (NB-IoT) is a cellular IoT technology designed for devices that send relatively small amounts of data, require wide-area coverage, and are expected to operate for long periods with low power consumption. It is particularly suitable for sensors and meters that do not need high data rates or continuous broadband connectivity. NB-IoT can use licensed cellular spectrum and is designed to provide reliable connectivity with efficient radio resource use.

The main design idea is to simplify the communication requirements of an IoT device while maintaining dependable wide-area connectivity. An NB-IoT endpoint normally contains a low-power sensor or controller, an NB-IoT radio modem and an antenna. Data is transferred through the cellular radio access network to the operator's core network and then to an IoT application platform or industrial control system.

2. NB-IoT Architecture

IoT Device / Sensor	NB-IoT Radio	Cellular RAN	Core Network	Application
Meter / Sensor Controller	Low-power Narrowband modem	eNB / LTE network	Packet core Connectivity	Cloud / IoT platform
Senses data	Transmits data	Provides coverage	Routes data	Monitoring & control

Basic NB-IoT data path from an industrial sensor to an IoT application.

Major Architectural Elements

- **NB-IoT device:** A sensor, meter or controller collects a small amount of information and prepares it for transmission.
- **Radio access network:** The cellular base station provides the wireless connection between the device and the operator network.
- **Core network:** Authentication, mobility/connectivity functions and packet-data transport connect the device to services.
- **IoT platform:** Receives, stores and analyses sensor information and can forward commands to industrial systems.
- **Application layer:** Converts the collected data into useful functions such as metering, alarms, asset monitoring or maintenance decisions.

3. Important NB-IoT Characteristics

NB-IoT is optimized for low-throughput IoT traffic rather than high-speed multimedia. Its narrowband operation, power-saving mechanisms and cellular coverage make it useful when devices transmit small packets periodically or only when an event occurs. The technology can support large numbers of low-complexity endpoints while reducing the need for local gateways in many deployments.

4. Emerging Industrial Applications

The strongest industrial use cases are those in which sensors are distributed over a large area, produce small data messages, and may be difficult or expensive to connect using wired infrastructure. NB-IoT is therefore well matched to monitoring applications where the communication requirement is more about coverage, reliability and battery life than high bandwidth.

Application	NB-IoT role	Industrial benefit
Smart metering	Electricity, gas and water meter data	Remote readings and reduced manual inspection
Asset monitoring	Periodic status/location information	Better visibility of distributed equipment
Environmental monitoring	Temperature, humidity, air/water quality sensors	Continuous remote condition monitoring
Predictive maintenance	Machine-condition sensor messages	Early detection of abnormal operating conditions
Smart agriculture	Soil and field sensors	Remote monitoring over wide areas
Industrial safety	Alarm and condition sensors	Remote alerts for abnormal conditions
Smart buildings	Occupancy/environment sensors	Energy and facility management

5. Evaluation and Critical Analysis

The major strength of NB-IoT is that it addresses a different problem from conventional mobile broadband. Industrial sensors frequently do not need high throughput, but they may need dependable coverage and long unattended operation. Using a cellular IoT technology can simplify wide-area connectivity because the operator network already provides radio infrastructure and managed connectivity.

However, NB-IoT should not be selected simply because it is a cellular technology. Applications that need high data rates, continuous real-time control, very low end-to-end latency, or large data streams may require other technologies. Industrial automation also has strict requirements for deterministic communication and safety. Consequently, NB-IoT is generally more appropriate for monitoring, telemetry, metering and event reporting than for replacing every real-time industrial control link.

Advantages

- Low-power operation supports long-lived battery-powered sensor deployments.
- Wide-area cellular coverage is useful for geographically distributed industrial assets.
- Small and infrequent data transfers match the traffic pattern of many sensors and meters.
- Licensed-spectrum operation can provide controlled and reliable wireless connectivity.
- Centralized cloud or IoT platforms simplify large-scale monitoring and data analysis.

Limitations

- Low throughput makes NB-IoT unsuitable for high-bandwidth industrial data.
- It is not a universal replacement for deterministic, ultra-low-latency industrial control networks.
- Coverage, operator availability and deployment cost must be considered for each industrial site.
- Battery life depends on message frequency, radio conditions and device configuration.

6. Industrial Data Flow and Deployment Model

A practical NB-IoT industrial deployment begins with sensors attached to machines, meters, tanks, pipelines, buildings or outdoor assets. The sensor periodically measures a parameter or detects an event. The NB-IoT modem then sends a compact message through the cellular network. The core network routes the data to an IoT platform, where it can be stored, visualized and analysed. If a threshold is crossed, the platform can generate an alert or send an appropriate command to the industrial system.

1. Sense	2. Process	3. Transmit	4. Analyse	5. Act
Sensor measures physical parameter	Controller prepares small data packet	NB-IoT → RAN → core network	IoT platform checks trends / thresholds	Alarm, report or maintenance action

Typical NB-IoT industrial monitoring workflow.

7. Evidence-Based Design Considerations

For a successful deployment, the communication design should begin with the application traffic profile. The engineer should estimate how frequently each sensor reports, the size of each message, the required coverage area, expected battery life and the consequence of a lost or delayed message. These requirements determine whether NB-IoT is appropriate. The system should also include suitable authentication, access control, secure data transport and application-level monitoring because industrial sensor data may influence maintenance and operational decisions.

8. Overall Assessment

NB-IoT provides an efficient connectivity layer for industrial IoT applications characterized by small data volumes, large geographic coverage and low-power operation. Its architecture connects simple field devices through a cellular radio access network and core network to centralized IoT services. The most promising industrial applications include smart metering, asset monitoring, environmental sensing, predictive-maintenance telemetry, agriculture and building management.

Its main value is not maximum data speed but an appropriate balance between coverage, device simplicity, power efficiency and managed connectivity. At the same time, engineers must recognize its limitations when applications demand high throughput or deterministic real-time control. Therefore, NB-IoT should be treated as a specialized connectivity choice within an industrial communication architecture rather than a universal replacement for every wireless technology.

9. Conclusion

NB-IoT architecture is well suited to emerging industrial applications that require reliable wide-area monitoring with low power consumption and modest data rates. A typical system consists of low-power sensors and controllers, NB-IoT radio connectivity, the cellular access and core network, and an IoT application platform. When the application requirements match these characteristics, NB-IoT can reduce manual monitoring, improve asset visibility and support data-driven maintenance and management. Its successful use depends on selecting it for the right traffic, latency, coverage and energy requirements.

Final architecture: Industrial Sensor → NB-IoT Device/Modem → Cellular RAN → Core Network → IoT Platform → Analytics/Alert → Industrial Decision or Maintenance Action.